

DASC7606C Group Project Proposal

SONIC-Based Language-Conditioned Tabletop Grasping, Visuomotor Learning, and Sim-to-Real Deployment for the Unitree G1

A Controlled Comparison of Kimodo and ARDY with Three Optional Extensions

Item	Description
Track	Track 4: Humanoid-Robot Motion Generation Deployment
Group size	12 students
Robot and environment	Unitree G1; the project will first extend SONIC's official G1 + MuJoCo sim-to-sim workflow with a tabletop grasping scene. If that extension proves infeasible, an adapted LIBERO/robosuite-MuJoCo setup will be evaluated. Real-robot experiments will only take place at Archon under supervision and after TA approval.
Required low-level controller	SONIC
Core motion-generation methods	Kimodo and ARDY
Core task	Language-driven grasping and lifting of a tabletop block
Three optional extensions	B1 Speech-to-Text; B2 Trainable vision-language motion model; B3 Sim-to-Real, with ACT as the first candidate and Isaac-GR00T as a conditional real-robot alternative, plus optional Qwen-based long-horizon task decomposition

Group Members

Fu Yuhan; Hao Hao; Huang Xinxuan; Li Jingyao; Luo Mingdi; Meng Guanlin; Mi Siyuan; Tang Zichun; Tu Chengyuan; Zhang Kunqi; Zhang Yixin; Zhang Yuanzhuo.

1. Project Summary

This project will build and evaluate a natural-language-controlled Unitree G1 tabletop manipulation system in simulation. The required baseline will compare two distinct pretrained motion-generation methods, Kimodo and ARDY. Both methods will share the same simulation scene, instruction set, motion adaptation interface, SONIC low-level controller, hand or gripper controller, and evaluation protocol. The target behavior is for the G1 to follow natural-language instructions and complete the full sequence of reaching toward a block, grasping it, and lifting it from a table.

In addition to the required baseline, the project proposes three optional extensions with clearly separated scopes and acceptance criteria:

- Bonus 1 - Speech-to-Text:** Convert spoken commands into the same textual instruction representation used by the baseline, while retaining text correction and fallback mechanisms.
- Bonus 2 - Trainable vision-language motion model:** Collect synchronized images, states, actions, and language data in G1 simulation; convert them into LeRobot episodes; train a model that maps head-camera images and text instructions to SONIC-compatible G1 upper-body motion; and evaluate generalization to unseen block positions.
- Bonus 3 - Sim-to-Real:** After completing simulation evidence, risk checks, and TA approval, first evaluate an ACT visuomotor baseline for real-robot deployment. Isaac-GR00T N1.7 will be treated as a conditional VLA alternative:

the team will first confirm the G1/SONIC action interface, legally usable task demonstrations, compute availability, and simulation performance before deciding whether to fine-tune it and request a supervised real-robot trial. For longer tasks such as moving objects, the team may also explore a local lightweight Qwen model, or an approved cloud API, to generate structured skill sequences that are verified and replanned step by step by an execution monitor.

All three bonuses are controlled by milestone gates. The text-input Kimodo/ARDY baseline will remain independently runnable, so none of the required Track 4 deliverables will depend on the completion of an optional extension.

2. Background and Motivation

Track 4 studies how natural-language instructions can control a Unitree G1 humanoid robot. The challenge extends beyond generating visually plausible human-like motion. The system must adapt generated motion to the G1 embodiment, execute it through SONIC, coordinate an independent hand or gripper controller, stabilize the lower body, and achieve reliable physical contact with an object.

Kimodo and ARDY provide two distinct motion-generation approaches suitable for controlled comparison. Running both methods with the same downstream controller and simulation scene allows the project to compare instruction following, grasp-pose accuracy, physical feasibility, response time, number of user corrections, and ease of integration.

The three extensions address three limitations of a fixed-scene, text-only baseline. First, speech commands provide a more natural interface but introduce recognition errors and additional latency. Second, a fixed text-to-motion model does not directly observe the current scene and may fail when block positions change. Third, simulation success alone does not demonstrate safe and stable transfer to a physical G1, and it does not address error accumulation or instruction tracking in long-horizon tasks such as moving multiple objects. The project will therefore examine speech input, vision-language motion learning, and a staged sim-to-real route with task decomposition and execution verification.

3. Project Scope and Alignment with Course Requirements

3.1 Required Core Scope

The core project will cover every mandatory Track 4 requirement:

- Participate in and document the supervised demonstration-recording and policy-inference workflows at Archon.
- Build a Unitree G1 simulation environment containing a table and a graspable block.
- Use SONIC as the low-level controller.
- Accept natural-language instructions entered by keyboard.
- Implement two distinct motion-generation methods: Kimodo and ARDY.
- Adapt both methods to the same G1/SONIC execution interface.
- Implement any required independent hand or gripper control.
- Demonstrate block grasping and lifting in simulation.
- Evaluate both methods with shared instructions, comparable initial conditions, repeated trials, explicit success criteria, and representative failures.
- Explain how the lower body is stabilized when only the upper body is controlled.

3.2 Optional Extensions

- **Bonus 1 - Speech interaction:** Speech command -> automatic speech recognition -> validated text command -> core or extended motion method.

- **Bonus 2 - Trainable vision-language motion learning:** Simulated head-camera image + text instruction + G1 state -> trainable motion generator -> SONIC-compatible G1 reference motion. Supporting work includes synchronized data collection, LeRobot conversion, and seen/unseen generalization evaluation.
- **Bonus 3 - Sim-to-Real and long-horizon execution:** In a supervised environment and after TA approval, first validate an ACT visuomotor baseline. If G1/SONIC data, compute, and simulation-validation conditions are satisfied, evaluate Isaac-GR00T N1.7 as an alternative VLA route. An optional Qwen high-level planner may convert long instructions into constrained skill sequences and perform a state check after each skill.

Each bonus will separately report its implementation scope, repeated experiments, failure cases, and reproducibility evidence. Only completed and evaluated components will be claimed as final contributions.

3.3 Scope Boundaries

- The required baseline will use pretrained Kimodo and ARDY checkpoints and will not require project-specific fine-tuning.
- Keyboard text remains the required reliable input. Speech is an additional layer and retains a text fallback.
- The first stable version will use a fixed robot initial pose, table position, and block position.
- Block-position randomization, multiple distinguishable blocks, dataset generation, and model training belong to the optional extensions.
- The project will first run SONIC's official Unitree G1 + MuJoCo sim-to-sim workflow, then add a table, block, camera, hand control, and grasp-success check. The official example validates the G1/SONIC control chain; the group must still implement and test the tabletop manipulation functions.
- If the official workflow cannot satisfy the tabletop grasping requirements before the feasibility gate, the team will evaluate the LIBERO/robosuite-MuJoCo option recommended by Jianyu or reuse relevant scene assets. Any LIBERO-based solution must still integrate G1 and SONIC. The original Panda + OSC configuration cannot substitute for the core experiment.
- The mandatory Archon session is intended to provide hands-on understanding of real-robot data collection and inference. The project does not depend on deploying the group's own system on the physical robot.
- Sim-to-real work is Bonus 3. Supervised deployment will only be attempted after the required simulation evidence, evaluation results, deployment plan, and safety checks have been reviewed and approved by the TAs.
- ACT or GR00T may not bypass SONIC and directly provide G1 whole-body low-level control. ACT outputs must be adapted and safety-checked before conversion into SONIC-compatible reference motion. Under the official G1/SONIC route, GR00T predicts SONIC latent motion tokens that are decoded by the matching SONIC control stack. The two routes use different action representations and cannot directly share one joint-trajectory adapter. Hand commands must be separately validated against the hardware and interface available at Archon.
- Qwen will only perform low-frequency, high-level task decomposition and will not enter the real-time motor-control loop. All outputs will be checked against a schema, a skill whitelist, and execution-state validation.

3.4 Simulation Platform Selection and Compliance Conditions

The course does not prescribe a particular simulator. The official SONIC project provides a G1 + MuJoCo sim-to-sim workflow, so this project will use it as the initial platform rather than starting with a LIBERO adaptation. Early experiments must determine whether the official workflow can be extended into a stable tabletop grasping task. The decision rules are:

1. Reproduce the official G1 + SONIC sim-to-sim example and confirm that the robot model, controller, reference motion, and simulator communication operate correctly.
2. Add a table, graspable block, camera, independent hand control, deterministic reset, and success check, then complete a scripted-grasp feasibility test.
3. If Step 2 cannot be stabilized before the scheduled gate, evaluate an adapted LIBERO/robosuite-MuJoCo setup or reuse its tabletop scene, object assets, language tasks, and camera configurations. If the complete LIBERO platform is used, its default Panda embodiment and OSC action path must be replaced with a G1 motion adapter

and SONIC.

- Regardless of scene source, compare Kimodo and ARDY with the same G1 model, SONIC configuration, camera, object, initial conditions, and success criteria. Record any difference between the simulator physics rate and SONIC control rate.

Official starting point: [SONIC sim-to-sim Quick Start](#).

3.5 Requirements Traceability

Track 4 requirement	Planned evidence
Archon session and workflow understanding	Participation records, permitted photos or notes, and an accurate report of inputs, recorded data, model outputs, control flow, safety procedures, and failure or recovery behavior
SONIC integration	Shared canonical motion interface, G1 adapter, SONIC configuration, control-flow diagram, tracking logs, and explanation of lower-body stabilization
Text-driven grasping	Demonstration of reach, grasp, and lift commands; step-by-step commands may be used when necessary
Two distinct methods	Independent Kimodo and ARDY implementations using the same SONIC controller and scene
Evaluation and analysis	Repeated trials, shared instructions and seeds, success rate, response time, correction count, precision, physical feasibility, and failure taxonomy
Demonstration and reproducibility	Successful and failed trials for both methods, public GitHub repository, configurations, setup instructions, and evaluation scripts
Report and presentation	A 5-8 page report excluding references; a 10-minute presentation with the required demonstration; and a 5-minute Q&A
Optional sim-to-real	Simulation videos, repeated evaluation, risk checklist, deployment plan, TA approval record, and evidence from successful and failed supervised trials

4. Project Objectives and Research Questions

4.1 Core Objectives

- Build a reproducible G1 tabletop simulation environment with SONIC as the shared low-level controller.
- Reproduce and deploy usable G1-oriented Kimodo and ARDY inference pipelines.
- Define a canonical motion representation so both methods can fairly connect to one G1/SONIC interface.
- Implement reliable grasp-stage control and any required independent hand or gripper control.
- Compare Kimodo and ARDY using matched instructions, initial conditions, compute budgets, and evaluation criteria.
- Produce clear demonstrations, failure analysis, and reproducible code and configurations.

4.2 Core Research Questions

- How do Kimodo and ARDY differ in instruction following and their ability to reach a specified grasp pose?
- Can spatial constraints or interactive text corrections improve grasp success?
- What trade-offs exist among motion quality, response time, physical feasibility, and ease of SONIC integration?
- Do failures originate from motion generation, embodiment adaptation, SONIC tracking, hand control, contact dynamics, or instruction understanding?

4.3 Optional Research Questions

- With what accuracy and latency can speech-to-text convert spoken manipulation commands into task-valid text instructions?
- Compared with keyboard text, how do speech-recognition errors affect correction count, total response time, and grasp success?

- Can a lightweight vision-language-conditioned motion generator produce SONIC-compatible grasp trajectories from a single head-camera image and a text instruction?
- Can the trained model generalize to block positions that were not included in the training set?
- If time permits, how do deterministic transformer regression and flow matching compare in motion quality, inference speed, and generalization?
- Can an ACT model trained and validated in simulation complete a fixed tabletop block grasp on the physical G1 while SONIC maintains stable and safe execution?
- If both routes obtain real-robot testing conditions, how do ACT and a fine-tuned GR00T model compare in success rate, response time, visual robustness, correction requirements, and sim-to-real degradation? If the test conditions do not match, results will be reported separately without a direct performance ranking.
- For long-horizon tasks such as moving objects, can a local Qwen task planner preserve step order, target identity, and completion state more reliably than a single small VLA/VLM policy?
- What trade-offs exist between local inference and an approved cloud API in planning quality, latency, reproducibility, privacy, and network dependence?

5. Overall System Design

5.1 Required Baseline Pipeline

```
Keyboard natural-language instruction
-> instruction validation and task parsing
-> Kimodo or ARDY
-> method-specific output adapter
-> canonical G1 reference motion
-> SONIC low-level controller
-> independent hand/gripper controller
-> simulated grasp and lift
```

5.2 Optional Speech-Input Pipeline

```
Speech command
-> speech-to-text
-> confidence check and command normalization
-> the same text interface used by the baseline
-> Kimodo or ARDY
-> G1 motion adapter
-> SONIC and hand/gripper control
```

When recognition confidence is too low or the parsed command is invalid, the system will ask the user to repeat the command or allow text correction. It will log the recognition transcript, confidence, latency, number of corrections, and final accepted instruction.

5.3 Optional Trainable Vision-Language Extension

```
Speech -----> speech-to-text -----> instruction L
                                     |
Head-camera image I -> vision-language-conditioned model ----+
                                     |
                                     v
                             upper-body trajectory Q_upper
                                     |
fixed standing reference -> complete G1 reference motion Q_ref
                                     |
                             SONIC + gripper control
                                     |
                             generalized grasp and lift
```

The extended model is defined as:

```
f_theta(I, L) -> Q_upper[1:T], where I is an H x W x 3 RGB image.
```

The lower body initially uses a fixed standing reference:

```
Q_lower[1:T] = Q_standing.
```

The complete motion is:

```
Q_ref[1:T] = [Q_lower[1:T], Q_upper[1:T]] in  $\mathbb{R}^{(T \times 29)}$ .
```

Reference joint velocity can be obtained by finite differences:

```
qdot_ref[t] = (q_ref[t] - q_ref[t-1]) / delta_t.
```

For stationary grasping, root position and orientation will use the standing reference unless the selected SONIC interface requires another representation. Hand or gripper commands will be represented as a separate sequence $G[1:T]$ and synchronized with the arm trajectory.

5.4 Optional Sim-to-Real and Long-Horizon Execution Pipeline

```
Long speech or text instruction
-> Speech-to-Text, if speech is used
-> Qwen high-level planner, local by default and approved API optional
-> constrained JSON skill sequence
-> ACT visuomotor baseline first, or fine-tuned Isaac-GR00T if conditions permit
-> route-specific action adaptation and safety checks
-> SONIC whole-body low-level control + validated hand/gripper control
-> visual/state completion check
-> continue, retry, recover, or replan
```

The initial real-robot scope is reach-grasp-lift with a fixed standing position, fixed table, and one known block. ACT is the first candidate because its action-chunking structure can predict a short segment of actions from images and robot state. It must connect to the language input through a task label or structured skill parameters and produce targets that can be safely executed through an adapter and SONIC. GR00T is an alternative VLA route; the proposal does not assume that its pretrained weights can directly perform the group's block-grasping task. A supervised physical test will only be considered after obtaining suitable G1/SONIC demonstrations, completing fine-tuning and simulation validation, and receiving approval from the TAs and Archon staff. If these conditions are not met, the project will limit this route to simulation feasibility analysis and will not list physical deployment as an expected deliverable.

The long-horizon planner will return a fixed schema such as:

```
{
  "steps": [
    {"skill": "locate", "object": "red_block"},
    {"skill": "reach", "object": "red_block"},
    {"skill": "grasp", "object": "red_block"},
    {"skill": "lift", "height_m": 0.10},
    {"skill": "move_to", "target": "right_container"},
    {"skill": "release"},
    {"skill": "verify", "condition": "red_block_in_right_container"}
  ]
}
```

The tentative local model is a quantized Qwen3-8B or a comparable lightweight model selected after hardware profiling. If an API is used, the project will record its model version, request template, latency, failure handling, and data policy. In either case, the LLM will invoke only whitelisted skills and will never output motor commands.

6. Functional Modules and Technical Approach

6.1 Module Overview

This section uses the following scope labels. **CORE** denotes functionality required for the Track 4 deliverables. **BONUS** denotes an optional extension that begins only after the core baseline is stable and cannot block the core submission. **CORE + BONUS** denotes shared infrastructure serving both the core experiments and completed extensions.

ID	Scope	Module	Main function	Output and acceptance criterion	Primary team
M1	CORE; position randomization, multiple blocks, and training-camera data are BONUS	Simulation platform, scene, and task definition	First reproduce the official SONIC G1 + MuJoCo sim-to-sim workflow; then add a table, block, camera, fixed initial state, reset, and success condition. If the grasping extension fails the gate, evaluate LIBERO/robosuite-MuJoCo adaptation or asset reuse.	Core: deterministic reset and fixed-scene grasping in the selected G1 + SONIC environment. Bonus: position randomization, multiple targets, and training observations.	Team 1
M2	CORE	Canonical motion interface and SONIC integration	Define the motion schema, map joint order and units, resample trajectories, calculate velocity references, configure standing/root references, and execute through SONIC.	Kimodo and ARDY both send comparable reference motion through the same interface.	Team 1, with Teams 2 and 3
M3	CORE	Grasping and hand control	Implement open, pre-grasp, approach, close, lift, release, and recovery states; synchronize hand commands with reference motion.	The robot grasps and lifts a block; hand/gripper behavior is logged separately.	Team 1
M4	CORE	Kimodo pipeline	Reproduce official inference, define inputs and constraints, generate motion, adapt outputs, and document limitations.	Reproducible Kimodo -> SONIC -> G1 grasping pipeline.	Team 2
M5	CORE	ARDY pipeline	Reproduce the official G1 configuration and inference, investigate interactive or streaming generation, adapt outputs, and document corrections.	Reproducible ARDY -> SONIC -> G1 grasping pipeline.	Team 3
M6	BONUS	Speech-to-text and instruction interface	Capture audio, run ASR, normalize commands, validate task syntax, handle confidence and retries, and provide a text fallback.	Speech reliably enters the shared text interface; transcript, confidence, latency, and corrections are logged.	Team 4
M7	BONUS	Simulation data collection and LeRobot export	Generate expert trajectories; record synchronized images, states, and actions; save instructions, timestamps, and episode boundaries; perform quality checks and convert to LeRobot.	Reusable dataset with scripts, schema, metadata, quality report, and an accessible dataset or link.	Team 1 leads; Team 4 performs quality control
M8	BONUS	Trainable vision-language motion model	Implement vision/text encoders, multimodal fusion, deterministic transformer trajectory decoder, loss, training, checkpoints, and SONIC inference adapter. Diffusion or flow matching is an optional later ablation.	The trained model generates SONIC-compatible upper-body trajectories and runs in the shared scene.	Teams 2 and 3
M9	BONUS	Sim-to-Real ACT/GR00T feasibility and deployment	Build a TA review package; calibrate real camera, state, and action interfaces; first validate ACT; evaluate the official fine-tuned GR00T G1/SONIC route only after data, compute, interface, and simulation gates pass.	Real-robot testing occurs only after approval. If conditions are insufficient, deliver simulation results, bottleneck analysis, and a deployment plan without claiming physical deployment.	Team 3 leads; Team 1 owns SONIC and safety interfaces
M10	BONUS	Qwen long-horizon planning and execution monitoring	Prefer a local Qwen3-8B or comparable model; optionally use an approved API; convert long commands into whitelisted JSON skills; verify each step, retry, and replan.	Complete at least one multi-step object-moving task in simulation; record step accuracy, replanning count, latency, and failure types.	Team 4 leads; Teams 2 and 3 expose skills
M11	CORE + BONUS	Evaluation, logging, and failure analysis	Evaluate the Kimodo/ARDY core and separately add STT, data quality, Bonus 2 vision-language model, generalization, sim-to-real, ACT/GR00T, and long-task evaluation.	A fair core comparison and separate results for completed bonuses, with simulation and real conditions clearly identified.	Team 4
M12	CORE	Archon, safety, documentation, and release	Record the supervised workflow, maintain setup instructions and contribution evidence, check licenses/data restrictions, and integrate the report, demonstration, and repository.	Accurate Archon section, public repository without restricted material, and complete contribution statement.	Team 4 leads; all members participate

6.2 Shared Motion Interface

To prevent incompatibility among pipelines, every motion method must export the following unified representation before entering SONIC:

```
CanonicalMotion
joint_names
```



```

q_ref[T, 29]
qdot_ref[T, 29] or a velocity calculation rule
timestamps[T]
root_position[T]
root_orientation[T]
optional_hand_command[T]
metadata: method, instruction, seed, object pose, checkpoint

```

The interface specification will define joint order, units, coordinate frames, sampling frequency, trajectory duration, missing-joint handling, clipping rules, and interpolation. Kimodo and ARDY may retain their internal representations, but both must pass through the same canonical contract before evaluation.

6.3 Simulation and Task State Machine

The platform will first use SONIC's official G1 + MuJoCo sim-to-sim workflow and build the tabletop task on top of it. Acceptance requires stable G1 reset, SONIC tracking of unified reference motion, independently controllable hand or gripper behavior, valid block contact and success checks, and identical conditions for Kimodo and ARDY. The official example does not provide all these grasping functions. If the tabletop extension cannot meet the early feasibility gate, the team will evaluate an adapted LIBERO/robosuite-MuJoCo setup or reuse its assets. Results from the original Panda + OSC configuration cannot serve as Track 4 core evidence.

The initial state machine is:

```

RESET
-> STAND
-> RECEIVE INSTRUCTION
-> GENERATE MOTION
-> PRE-GRASP
-> APPROACH
-> CLOSE HAND/GRIPPER
-> LIFT
-> VERIFY SUCCESS
-> END or RECOVER

```

Core experiments will fix the robot initial pose, table position, block position, and camera pose. Once both baseline methods are stable, the extension will randomize block positions within a defined reachable region. If the scene contains multiple blocks, instructions will include color or identity information so that language conditioning is meaningful.

6.4 Kimodo Method

The Kimodo team will:

- Reproduce a usable G1-oriented pretrained inference example.
- Record text input, kinematic constraints, motion representation, checkpoint, preprocessing, and compute requirements.
- Generate reference motion for reaching, grasp approach, and lifting.
- Investigate whether hand or end-effector constraints derived from object pose improve accuracy.
- Map Kimodo output to the canonical motion contract.
- Connect to the shared SONIC and gripper interfaces.
- Record method-specific failures, including unsatisfied constraints, trajectory discontinuities, joint-limit violations, and contact failures.

6.5 ARDY Method

The ARDY team will:

- Reproduce the pretrained G1 configuration and official inference example.
- Record model inputs, outputs, checkpoints, preprocessing, and compute requirements.
- Investigate interactive or streaming generation where technically feasible.

- Implement corresponding reaching, grasp-approach, and lifting behaviors under the same task definition.
- Map ARDY output to the canonical motion contract.
- Connect to the same SONIC and gripper interfaces.
- Record method-specific failures and the effect of interactive corrections.

6.6 Optional Speech-to-Text Interface

The speech module will not directly control the robot. It will create validated text commands for the baseline interface. The initial command grammar will support target identity and manipulation phase, for example:

```
Pick up the red block.
Reach toward the blue block.
Adjust the right hand above the block.
Grasp the block.
Lift the block.
```

The module includes:

- Audio capture with explicit start and stop controls.
- Selection of an offline or online ASR engine based on latency, privacy, and deployment tests.
- Normalization of object names, colors, and action verbs.
- Confidence thresholds and rejection of out-of-scope commands.
- Confirmation or retry for low-confidence output.
- Text correction and text fallback.
- Logging of raw transcript, normalized command, confidence, latency, and number of corrections.
- A privacy rule that prohibits publishing participant voice recordings without permission.

6.7 Optional Simulation Data Collection

Each training episode will:

1. Reset the G1 to a fixed standing pose.
2. Sample a block position within the reachable tabletop area.
3. Render a head-camera RGB image.
4. Assign a keyboard instruction or speech-recognized instruction.
5. Generate an expert pre-grasp, grasp, and lift trajectory using scripted inverse kinematics or a motion planner.
6. Execute the trajectory in simulation.
7. Verify success and label the failure reason when applicable.
8. Save synchronized observations, robot state, action, joint reference, hand command, timestamp, and episode boundary.
9. Convert qualified episodes into LeRobot format.

A compact training item is:

```
D_i = (I_i, L_i, Q_i_GT, G_i_GT).
```

The published episode representation will also contain synchronized robot states, actions, timestamps, camera metadata, success labels, and optional audio/transcript metadata.

Scripted IK or motion planning is the first expert source because it provides explicit geometric relationships, accurate grasp targets, low data-generation cost, and less contamination from pretrained-model errors. High-quality Kimodo or ARDY rollouts may be included as additional demonstrations but will be separately labeled.

6.8 Optional Trainable Vision-Language Motion Generator

The proposed lightweight architecture is:

```
Pretrained Vision Encoder
+ Lightweight Text Encoder
+ Small Multimodal Fusion Transformer
+ Motion Decoder
```

Component	Initial configuration
Vision encoder	DINOv2 ViT-S/14 or another lightweight pretrained ViT; initially frozen; retain spatial patch tokens
Text encoder	Pretrained MiniLM-class encoder; initially frozen
Shared hidden dimension	256
Fusion transformer	4 layers, 8 attention heads, FFN dimension 1024
Estimated trainable parameters	Approximately 5-10M, excluding frozen encoders
Primary decoder	Deterministic transformer trajectory regression
Optional decoder	Flow matching or DiT-style motion generator after the deterministic baseline is stable

The representation pipeline is:

```
I -> Z_v,
L -> Z_l,
(Z_v, Z_l) -> Z_condition.
```

The deterministic decoder predicts:

```
(Z_v, Z_l) -> Q_hat[1:T].
```

The initial loss is:

```
L_pos = (1/T) * sum_t ||q_hat[t] - q_GT[t]||_1

L_vel = (1/(T-1)) * sum_(t=2..T)
        ||(q_hat[t] - q_hat[t-1]) - (q_GT[t] - q_GT[t-1])||_1

L = L_pos + lambda * L_vel.
```

If the deterministic baseline is reliable, an optional flow-matching decoder will learn:

```
v_theta(Q_tau, tau, Z_condition),
dQ_tau/dtau = v_theta(Q_tau, tau, Z_condition),
```

starting from $Q_-(\tau=0) \sim N(0, I)$ and generating $Q_-(\tau=1) = Q_{ref}[1:T]$.

6.9 Optional Generalization Design

Training and test sets will be separated by block position. For a reachable tabletop area:

```
x in [x_min, x_max],
y in [y_min, y_max],
P_train intersection P_test = empty set.
```

The primary generalization comparison will separately report performance on seen and unseen positions. If multiple colored blocks are used, target identity and dataset splits will be recorded so target-selection accuracy can be calculated without data leakage.

6.10 Optional Sim-to-Real: ACT First, Isaac-GR00T as an Alternative

Sim-to-real development uses staged gates:

- S0 - Freeze simulation:** Freeze the G1/SONIC version, camera pose, state/action schema, action limits, gripper logic, and success criteria.
- S1 - Safety validation:** Test joint limits, velocity and acceleration limits, self-collision and table collision, abnormal-input rejection, emergency stop, and recovery.

3. **S2 - TA review:** Submit simulation videos, repeated-trial results, representative failures, and a deployment plan. Work remains in simulation without explicit approval.
4. **S3 - Supervised ACT test:** After approval, perform no-object slow-motion and reach tests before attempting single-block grasp-and-lift.
5. **S4 - GR00T alternative-route review:** Only when the team has approved G1/SONIC task demonstrations, sufficient fine-tuning compute, compatible hand hardware and camera configuration, and successful functional and safety validation in simulation will it submit a separate GR00T real-robot test plan. Supervised testing will only occur after approval.
6. **S5 - Result analysis:** Separately report simulation-to-real changes in success rate, action latency, visual failures, grasp failures, and human safety interventions. ACT and GR00T will only be compared directly when task, hardware, cameras, trial counts, and safety boundaries are comparable.

The tentative ACT input/output contract is:

```
inputs:
  head_rgb[t]
  robot_state[t]
  task_or_skill_condition
outputs:
  action_chunk[t:t+H]
    upper-body / end-effector target
  gripper target
```

ACT action chunks must pass through coordinate conversion, joint and velocity limits, temporal resampling, and reachability checks before conversion into SONIC-compatible references. Because original LIBERO data uses the Panda embodiment, a LIBERO ACT checkpoint cannot be deployed directly on G1. Visual representations may be pretrained, but the G1 action head must be trained or adapted using G1-specific demonstrations.

The official Isaac-GR00T N1.7 G1/SONIC route differs from the ACT joint-trajectory route. It collects camera images, robot state, language task descriptions, and demonstrated actions; organizes the data with a G1/SONIC embodiment configuration; and fine-tunes the VLA on matching demonstrations. The fine-tuned model predicts SONIC latent motion tokens and corresponding hand actions, which the SONIC stack executes. The official example represents action using a 64-dimensional motion token plus left- and right-hand actions. This project must confirm the actual G1 hand configuration and interface available at Archon and will not assume direct compatibility. The official fine-tuning tutorial recommends multi-GPU training resources, specifically four or more GPUs, so the team must profile its available compute and training cost. Public GR00T weights do not make the group's block-grasping task immediately usable; data, interface, compute, and simulation validation are required first.

This route is a conditional Bonus 3 alternative. It does not replace the mandatory Kimodo/ARDY + SONIC simulation comparison, and it does not assume that the mandatory Archon session permits the collection or removal of training data. If the team only completes interface investigation, simulation inference, or a deployment design, the final report will state that actual scope and will not describe it as a real-robot result.

Technical references: [Isaac-GR00T repository](#), [NVIDIA G1 + SONIC data collection, fine-tuning, and deployment workflow](#), and [VLA inference on G1](#).

6.11 Optional Qwen Long-Horizon Decomposition and Closed-Loop Execution

The Qwen layer addresses high-level step tracking rather than low-level motion generation. It receives a normalized text instruction and an optional structured scene summary, then returns a fixed JSON schema. A validator rejects unknown skills, missing parameters, targets outside the workspace, and excessively long plans. The executor sequentially invokes whitelisted skills such as `locate`, `reach`, `grasp`, `lift`, `move_to`, `release`, and `verify`.

Each step returns:

```
skill_id, status, confidence, observed_state,
failure_code, retry_count, elapsed_time
```

A dependent step can begin only after `verify` passes. On failure, the system performs a limited number of retries, returns to a safe standing pose, or requests replanning according to a predefined policy. A cloud API is an optional planning backend and will not perform real-time control. The local model is the default to improve reproducibility and

reduce network, privacy, and availability risks. If the planner receives text only, it has no direct access to visual state; a VLM or detector must provide a structured scene summary and skill-completion signals.

7. Supervised Archon Visit Plan

The group will book the visit at least two days in advance and participate under supervision. The report will accurately record:

- Robot and sensor configuration.
- Inputs and operator interface for demonstration recording.
- Recorded observations, robot state, actions, timestamps, and episode boundaries.
- Data storage and quality-control workflow.
- Policy-inference inputs and preprocessing.
- Model outputs and their representation.
- Conversion from model outputs to robot-control commands.
- Low-level controller, hand/gripper branch, safety filters, emergency stop, and recovery workflow.
- At least one successful workflow and one failure mode observed directly or explained by staff.
- Similarities and differences between the Archon workflow and the group's simulation/SONIC pipeline.

The group will retain only permitted notes, images, videos, and unrestricted metadata. Credentials, restricted robot data, internal code, and third-party material will not be published without permission.

8. Evaluation Plan

8.1 Experimental Controls

The Kimodo/ARDY core comparison will use:

- The same G1 model, simulator, scene assets, table, and object geometry.
- The same SONIC version, controller configuration, standing reference, and gripper logic.
- The same instruction semantics and grasp-success definition.
- Matched initial conditions and evaluation seeds.
- The same trajectory duration and resampling rules where applicable.
- Comparable compute and inference budgets.
- The same number of repeated trials per condition.

After pilot testing, the target is at least 20 trials for every reported condition. If runtime or compute constraints require a different number, all compared methods will still use the same number of trials, and the report will state the change.

8.2 Core Comparison Matrix

Factor	Core values	Extension values
Motion method	Kimodo, ARDY	Trainable vision-language generator when supported
Instruction input	Keyboard text	Speech-to-text with text correction
Scene	Fixed robot, table, camera, and block position	Randomized block positions; optionally multiple distinguishable blocks
Controller	Same SONIC configuration	Same SONIC configuration

Factor	Core values	Extension values
Evaluation seeds	Shared held-out seeds	Separate seen and unseen position sets
Embodiment/domain	Simulated G1	Approved physical G1; ACT first and GR00T as a conditional alternative
Task horizon	One reach-grasp-lift task	Qwen-decomposed multi-step object-moving task

8.3 Evaluation Metrics

Metric	Definition or measurement	Scope
Grasp success rate	Percentage of trials in which the target block is grasped and lifted above a specified height within the time limit	All motion methods
Target-selection accuracy	Percentage of multi-object trials in which the instructed target is grasped	Language/vision extensions
Instruction-following accuracy	Whether the specified target and action sequence are completed	All methods
Response time	Time from command submission to completion of motion generation; ASR and motion-generation latency are also reported separately	All methods and STT
User corrections	Number of repeated or edited instructions before execution is accepted	Text and speech input
Speech-recognition quality	Word error rate and task-level intent/slot accuracy for target and action words	STT extension
Motion precision	Euclidean end-effector or grasp-pose error where measurable: $E_{EE} = \text{distance}(p_{EE}, p_{\text{target}})$	All motion methods
SONIC tracking quality	Reference-to-executed joint error and number of tracking failures where available	All motion methods
Physical feasibility	Joint-limit violations, collisions, unstable motion, foot/root instability, and infeasible contact behavior	All motion methods
Generalization	Seen-position and unseen-position success rates on separated position sets	Trainable extension
Sim-to-real degradation	Change in success rate, precision, and latency for the same task from simulation to the physical robot	Bonus 3
Long-task completion	Percentage of trials completing all required skills with a verified final state	Qwen/long-task extension
Planning reliability	Invalid-plan rate, correct skill ordering, replanning count, and planner latency	Qwen/long-task extension

Metric	Definition or measurement	Scope
Safety intervention	Number of emergency stops, human aborts, safety-filter rejections, and recovery events	All approved real-robot experiments
Failure analysis	Counts for perception, ASR, instruction, generation, adaptation, controller, grasp/contact, and stability failures	Entire system

8.4 Planned Analyses

- Kimodo versus ARDY under fixed conditions with keyboard text input.
- Keyboard text versus speech-recognized text for each stable method.
- Performance before and after spatial constraints or interactive correction, where supported.
- Seen versus unseen block positions for the trainable extension.
- Vision-only versus vision-language conditioning when the data supports multiple targets.
- Global visual embedding versus spatial patch tokens when compute permits.
- Frozen versus partially fine-tuned visual backbone when compute permits.
- Deterministic regression versus flow matching only after the deterministic training baseline is complete.
- ACT performance in matched simulation and real-robot tasks.
- ACT versus fine-tuned GR00T only when both real-robot tests are approved and conditions are comparable, with cameras, action spaces, trial budgets, and safety boundaries stated explicitly. Otherwise, results will be presented separately without direct ranking.
- Fixed skill sequence versus local Qwen planning; if API use is approved, compare local and API planning accuracy, latency, and recovery.

A high success rate alone will not be treated as sufficient evidence of high-quality work. The analysis will emphasize fair conditions, repeated experiments, uncertainty, integration trade-offs, representative failures, and technically supported interpretation.

9. Team Structure and Responsibilities

The group will be organized by functional module rather than as two independent end-to-end method teams. The four functional teams will be led by Meng Guanlin, Hao Hao, Li Jingyao, and Zhang Yixin. Shared infrastructure will be implemented once and exposed through defined interfaces to Kimodo, ARDY, and the optional trainable model. Each team must complete its **CORE** responsibilities before beginning its **BONUS** responsibilities.

9.1 Functional Teams

Team	Lead and members	Core responsibilities	Bonus responsibilities after core completion
Team 1: Simulation, SONIC, and Grasp Control	Lead: Meng Guanlin; Fu Yuhan; Luo Mingdi	Reproduce the official G1 + SONIC simulation workflow; conduct tabletop grasping feasibility tests; define the canonical motion contract; implement standing/lower-body stabilization and hand/gripper control; evaluate LIBERO only if required	Position randomization, expert rollouts, LeRobot data, and the real-robot SONIC/safety adapter
Team 2: Kimodo	Lead: Hao Hao; Tang Zichun; Zhang Kunqi	Kimodo deployment, text/kinematic constraints, G1 output adaptation, shared SONIC interface, method-specific tests, and failure analysis	Vision/text encoding, multimodal fusion, data interface, and support for the Qwen skill-condition interface

Team	Lead and members	Core responsibilities	Bonus responsibilities after core completion
Team 3: ARDY	Lead: Li Jingyao; Huang Xinxuan; Zhang Yuanzhuo	ARDY deployment, streaming/interactive behavior, constraints, G1 output adaptation, shared SONIC interface, method-specific tests, and failure analysis	Trajectory decoder, ACT baseline, GR00T/G1-SONIC data and interface feasibility, and approved sim-to-real implementation
Team 4: Evaluation, Speech, and Release	Lead: Zhang Yixin; Mi Siyuan; Tu Chengyuan	Experimental design, shared seeds/logs, metrics, failure taxonomy, statistical summaries, video evidence, report/presentation integration, and repository reproducibility	Speech-to-text, Qwen planner/validator, evaluation of all three bonuses, TA deployment evidence package, and real-robot safety-event records

9.2 Tentative Individual Responsibilities

The following allocation defines initial ownership. The final contribution statement will reflect completed work and supporting evidence.

Member	Initial responsibility
Meng Guanlin	Team 1 lead: shared simulation/SONIC architecture, lower-body stabilization strategy, task state machine, and team coordination
Fu Yuhan	CORE: canonical motion interface, hand/gripper control, scripted grasp, and cross-team integration. BONUS: expert/data interface and real-robot SONIC motion-safety adapter
Luo Mingdi	CORE: table/block scene, camera setup, fixed reset, and observation hooks in the official G1 + SONIC simulation; evaluate LIBERO assets or platform adaptation if the extension is blocked. BONUS: position randomization, training-data hooks, and sim-to-real camera/calibration support
Hao Hao	Team 2 lead: Kimodo technical route, constraints, canonical-motion adapter, and SONIC integration coordination
Tang Zichun	CORE: Kimodo inference reproduction, output adaptation, and interface testing. BONUS: multimodal fusion and structured skill-conditioning support
Zhang Kunqi	CORE: Kimodo-specific tests, failure records, and reproducibility documentation. BONUS: visual/text encoders, data interface, and planner-to-policy interface testing
Li Jingyao	Team 3 lead: ARDY technical route, official inference reproduction, shared interface, and technical coordination of ACT/GR00T real-robot alternatives
Huang Xinxuan	CORE: ARDY deployment, constraints, method-specific tests, and failure analysis. BONUS: trajectory decoder and ACT baseline training/deployment
Zhang Yuanzhuo	CORE: ARDY streaming/interactive interface, canonical-motion adapter, and SONIC debugging. BONUS: GR00T G1/SONIC embodiment and latent-token inference validation, ACT action-chunk adapter, and approved real-robot interface tests
Mi Siyuan	CORE: shared logs and experimental records. BONUS: speech-to-text, command normalization, Qwen input interface, confidence/retry logic, and STT evaluation
Zhang Yixin	Team 4 lead: core evaluation protocol, results tables, statistical summaries, cross-team experiment coordination, gates for all three bonuses, and TA deployment evidence package
Tu Chengyuan	CORE: grasp-success checker, SONIC tracking metrics, failure taxonomy, videos, and repository reproducibility. BONUS: Qwen schema validator, long-task/sim-to-real metrics, and safety-event records

9.3 Shared Engineering Rules

- The entire group will use the same simulator version, G1 model, SONIC revision, coordinate convention, and configuration format.
- Interface documentation will be completed before independent method development begins.
- Every feature must have an owner, reviewer, test case, configuration, and evidence link.
- Shared interfaces and evaluation code will be reviewed through pull requests or an equivalent process.
- Every weekly integration check will run both methods through the current shared pipeline.
- Individual work logs will record tasks, commits or files, experiments, results, reviews, and meeting decisions.

10. Work Plan and Milestones

Milestone	Main work	Completion criterion
M0: Requirements and Archon workflow	Confirm course scope, attend the Archon session, record the data-collection/inference workflow, and freeze the initial interface plan	Scope record confirmed; Archon workflow documented; safety and publication restrictions recorded
M1: Shared platform	First run the official G1 + SONIC sim-to-sim workflow; then add table/block assets, scripted grasp, canonical motion contract, and gripper state machine. If the grasp extension fails the gate, evaluate LIBERO adaptation or asset reuse.	Platform decision documented; one scripted reference motion reproducibly grasps and lifts a block through SONIC
M2: Independent method baselines	Reproduce Kimodo and ARDY inference and map outputs into the canonical representation	Official examples reproduced; both methods generate inspectable G1 reference motion
M3: Core end-to-end integration	Use keyboard instructions to run Kimodo and ARDY through the shared SONIC/gripper pipeline	Both required pipelines complete at least one end-to-end grasp-and-lift trial
M4: Controlled core evaluation	Freeze instructions, seeds, success criteria, trial budget, and logging; collect successful and failed trials	Fair repeated comparison with complete logs, metrics, videos, and failure labels
M5: BONUS speech extension	Add audio capture, ASR, command normalization, confidence handling, text fallback, and latency/error logging	Speech controls at least one stable baseline method; STT accuracy and latency reported
M6: BONUS data and training extension	Automatically collect simulation data, export LeRobot episodes, and train the selected deterministic vision-language trajectory model	Dataset and quality checks documented; trained checkpoint produced; inference works through the shared adapter
M7: BONUS generalization and optional ablations	Evaluate separated unseen positions and selected ablations; attempt diffusion/flow matching only after prior work is complete	Seen/unseen results, representative failures, and evidence for completed ablations
M8: BONUS sim-to-real approval and ACT	Freeze the simulation configuration, complete safety checks, and submit the TA review package; after approval, test ACT in stages from slow motion to reach to grasp	If explicitly approved, record supervised tests, sim/real differences, failures, and safety interventions; otherwise retain simulation results
M9: BONUS GR00T alternative and long-horizon planning	Check G1/SONIC demonstrations, hand interface, and compute; after fine-tuning and simulation validation, separately apply for a supervised GR00T test; deploy the local Qwen planner in simulation and optionally compare an approved API backend	Report the actual GR00T stage achieved and its limitations; provide independent results and safety records if real testing occurs; report only validated long-task results

Milestone	Main work	Completion criterion
M10: Final materials	Complete the public GitHub repository, report, presentation, demonstration videos, contribution statement, and LLM/external-resource disclosure	Reproducibility review passed; evidence linked for every required and completed optional component

11. Deliverables and Reproducibility

11.1 Required Deliverables

- **Report:** 5-8 pages excluding references, covering the Archon session, simulation system, Kimodo and ARDY, SONIC integration, evaluation, failure analysis, and completed extensions.
- **Presentation:** 10 minutes including a required demonstration, followed by 5 minutes of Q&A.
- **Videos:** Text instructions and corresponding robot behavior, including successful and failed trials for both Kimodo and ARDY; extension evidence will be shown separately.
- **Public GitHub repository:** Implementation, configuration, setup instructions, evaluation code, and reproduction guide.
- **Contribution statement:** Description of each member's actual work with supporting evidence.

11.2 Optional Extension Deliverables

- Speech-to-text interface and command-normalization code.
- Anonymized or consented voice-test inputs when publication is permitted.
- Simulation recording and LeRobot conversion scripts.
- Dataset or accessible link when publication is permitted.
- Model training/inference code, configurations, and checkpoints.
- Seen/unseen split definitions and evaluation scripts.
- ACT training/inference configuration, G1 action schema, and sim-to-real adapter; if conditions permit, GR00T fine-tuning configuration, G1/SONIC embodiment mapping, inference configuration, and simulation-validation results.
- Qwen planner prompt/schema, whitelisted skill definitions, validator, execution monitor, and long-task logs.
- TA review summary, real-robot configuration, sim-to-real metrics, and safety-event statistics where publication is permitted.
- Successful and failed extension videos.
- Ablation results for completed optional comparisons.

11.3 Repository Structure

```

configs/
docs/
  archon_workflow/
  interfaces/
simulation/
  libero_adapter/
sonic_adapter/
gripper_control/
methods/
  kimodo/
  ardy/
  act/
speech_interface/
data_collection/
lerobot_conversion/
bonus_model/
task_planner/
  qwen/
sim2real/
evaluation/
scripts/
videos_or_links/

```

The repository will pin dependencies where practical, document hardware/software requirements, provide sample inputs and a small test configuration, and exclude credentials, restricted robot data, and third-party assets that cannot be legally published.

12. Feasibility, Risks, and Mitigations

Risk	Potential impact	Mitigation
Kimodo/ARDY representation mismatch	Generated motion cannot fairly connect to SONIC	Freeze the canonical motion contract early; validate joint order, units, coordinate frames, frequency, and trajectory continuity with small tests
Unstable SONIC or G1 integration	Tracking failure, balance problems, or unsafe motion	Begin with a fixed standing pose and upper-body motion; limit joint ranges; validate slow scripted motion before generated trajectories
Unstable grasp/contact	Visually plausible motion fails to contact, close, or lift reliably	Use explicit pre-grasp, grasp, and lift states; precise object pose; independent hand control; and contact-aware failure logs
Official SONIC simulation lacks a ready tabletop task	Scene, hand control, or contact-check extensions require more work than expected	Complete the scripted-grasp gate early; if it fails, evaluate LIBERO/robosuite-MuJoCo adaptation or asset reuse while preserving the G1 + SONIC control path
LIBERO embodiment/controller mismatch	As a backup, its default Panda + OSC configuration cannot directly support G1 + SONIC	Use the full LIBERO platform only after G1, SONIC, and shared evaluation are adapted; otherwise reuse only suitable assets
Incompatible parallel implementation	Teams cannot share one pipeline	Use shared configurations, interface tests, weekly integration runs, and cross-team code review
STT recognition error	Wrong target/action, more corrections, or an unsafe command	Restrict command grammar, use confidence thresholds, confirm low-confidence results, reject invalid commands, and keep text fallback
STT service latency or privacy issue	Slow interaction or voice data that cannot be published	Compare offline and online options, report latency by component, obtain consent, and do not publish voice data without permission
Poor training-data quality	Model learns inaccurate or unstable trajectories	Prefer scripted IK/planner experts; automatically check success; inspect samples; log failures; report dataset statistics
Train/test leakage	Inflated generalization results	Define disjoint position sets before training, save split manifests, and do not tune on test positions

Risk	Potential impact	Mitigation
Compute limitations	Training or large evaluation cannot be completed	Initially freeze encoders, use a small fusion model, prioritize deterministic regression, profile early, and attempt flow matching only after the baseline
ACT/GR00T sim-to-real domain gap	Simulation succeeds but the real robot fails because of camera, dynamics, or contact differences	Use approved G1-specific demonstrations, calibrate camera/state/action interfaces, apply domain randomization, test in slow stages, and enforce strict fallback conditions
ACT/GR00T action-space or hand-hardware incompatibility	Output cannot connect to SONIC or control the available hand	Separately validate the ACT reference-motion adapter and GR00T latent-token route; check real hand degrees of freedom and control interface; reject out-of-range or abnormal actions
Insufficient GR00T data or compute	Reliable fine-tuning or project-schedule simulation validation is impossible	Check approved demonstrations, training resources, and task match early; stop the real-robot alternative if conditions are insufficient and report feasibility findings only
Incorrect or infeasible Qwen plan	Wrong skill order, target confusion, or long-task failure	Use a fixed JSON schema, whitelisted skills, parameter and workspace checks, stepwise verification, limited retries, and safe recovery
Cloud API latency, outage, or data-policy risk	Poor reproducibility, interrupted control, or sensitive-data exposure	Use local Qwen by default; use an API only for approved, non-real-time high-level planning; pin model version; apply timeouts/fallback; minimize uploaded data
Scope expansion	Bonus work delays the required Kimodo/ARDY comparison	Use milestone gates, keep the text baseline independently runnable, and pause extensions whenever core deliverables are at risk
Restricted Archon or third-party data	Academic-integrity, safety, or licensing violation	Obtain explicit permission, record restrictions, publish only permitted content, and verify all external resources

13. Safety, Data Management, and Academic Integrity

- All physical-robot interaction will take place at Archon under supervision and laboratory safety procedures.
- Before any optional deployment plan is submitted, simulated motion will be checked for joint limits, collisions, instability, and abnormal commands.
- ACT, GR00T, and Qwen outputs will pass through interface checks, action bounds, and a safety state machine appropriate to each representation. An LLM or API will never directly output or transmit motor commands.
- Physical deployment will proceed from slow motion to no-object reaching and finally single-object grasping. Every stage permits human abort and return to a safe standing pose.
- Voice recording requires informed consent. Privacy-sensitive tests can always use text instructions.
- The public repository will not contain credentials, restricted robot data, unpublished internal material, or unauthorized third-party assets.
- Sources and licenses will be recorded for external code, models, checkpoints, datasets, and assets.
- Experimental failures and negative results will be reported honestly. Video selection will not conceal representative failure modes.
- Any sim-to-real attempt requires simulation evidence and a deployment plan submitted to the TAs in advance, followed by explicit approval and supervised booking.

14. LLM Usage Statement

ChatGPT and Codex were used during the preparation of this proposal.

- **Sections 1–4 — Project scope, objectives, and research questions:** The tools were used to organize the Track 4 requirements, clarify the distinction between required tasks and optional extensions, and improve the structure of the proposed research questions. The group reviewed these sections against the official project brief and made the final decisions on project scope.
- **Sections 5–8 — System design, technical approach, and evaluation plan:** The tools assisted in comparing candidate technical routes, organizing the Kimodo, ARDY, SONIC, ACT, Isaac-GR00T, and LIBERO components, and drafting descriptions of interfaces, evaluation metrics, risks, and experimental controls. Technical claims and external references were checked against the official repositories and documentation by the group. The proposed architecture and method selection were discussed and approved by the group members.
- **Sections 9–13 and 15 — Team organization, milestones, deliverables, risks, and expected contributions:** The tools were used to help structure responsibilities, milestones, risk-mitigation measures, and deliverables. The actual allocation of responsibilities and project priorities was determined through group discussion.
- **English translation and language editing:** Codex was used to translate the Chinese proposal into English and improve grammar, clarity, consistency, and formatting. Group members will review the English version before submission to ensure that it accurately represents the intended project plan.

No experimental results, datasets, robot demonstrations, or performance measurements in this proposal were generated or fabricated by an LLM. Any technical details that remain tentative are identified as proposed plans or optional extensions.

The group remains responsible for all submitted content. Before submission, group members will review and verify the technical claims, references, model descriptions, equations, and proposed implementation details. Any subsequent use of LLMs for code generation, debugging, experiment analysis, figure preparation, or report writing will be recorded and disclosed in the final report, together with the affected files, sections, or system components.

15. Expected Contributions and Final Scope

The required contribution is a controlled and reproducible comparison of Kimodo and ARDY for Unitree G1 tabletop grasping in simulation. Both methods will use the same SONIC low-level controller, hand/gripper logic, instructions, experimental conditions, metrics, and trial budget. The project will document the complete motion-to-control pipeline and analyze both successful trials and failures in integration and physical interaction.

The three optional extensions add a natural speech interface, a trainable perception-to-motion route, and approved staged sim-to-real/long-horizon execution. Isaac-GR00T is a conditional physical-robot alternative within the third extension:

Bonus 1: Speech -> Speech-to-Text -> validated text instruction

Bonus 2: Image + text + G1 state
 -> trainable vision-language motion model
 -> SONIC-compatible reference

Bonus 3: Long instruction -> optional Qwen structured skill plan
 -> ACT first, or fine-tuned GR00T if conditions permit
 -> route-specific safety checks -> SONIC + hand/gripper
 -> stepwise visual/state verification -> retry or replan

The extension plan covers speech-conditioned interaction, synchronized G1 simulation data collection, LeRobot export, trainable vision-language trajectory generation, unseen-position generalization, approved ACT real-robot testing, a conditional GR00T/G1-SONIC alternative, and Qwen-supported long-task decomposition with closed-loop verification. The baseline comparison remains the priority. Only extension components that are implemented, evaluated through repeated experiments, compliant with TA and safety requirements, and supported by reproducible evidence will be claimed as final project contributions.